Question 1 (1 point)
An AR(1) process is Y(t) = 3 + 0.5Y(t-1) + e(t). What is the unconditional mean of the process?

Question 1 options:

3

0.5

0

6

To find the unconditional mean (μ) of an AR(1) process given by:

$$Y(t) = 3 + 0.5Y(t-1) + e(t)$$

We follow these steps:

Step 1: The formula for the unconditional mean of an AR(1) process is:

$$\mu = \frac{c}{1-\phi}$$

Where:

- c is the constant term (3 in this case).
- ϕ is the coefficient of $Y(t-1)$ (0.5 in this case).

Step 2: Plug the values into the formula:

$$\mu = \frac{3}{1-0.5} = \frac{3}{0.5} = 6$$

Thus, the unconditional mean of the process is **6**.

The correct answer is: **6**.

Question 2 (1 point)
An AR(1) process is Y(t) = 2 + 0.5Y(t-1) + e(t). What is the unconditional variance of the process if variance of the error term is 0.5?

Question 2 options:

0.5

0.6667

2

4

To calculate the unconditional variance of an AR(1) process, we use the following formula:

$$\sigma^2 = \frac{\mathrm{Var}(e_t)}{1 - \phi^2}$$

Where:

- $\mathrm{Var}(e_t)$ is the variance of the error term (0.5 in this case),
- ϕ is the coefficient of $Y(t-1)$ (0.5 in this case).

Step 1: Plug the values into the formula:

$$\sigma^2 = \frac{0.5}{1 - (0.5)^2} = \frac{0.5}{1 - 0.25} = \frac{0.5}{0.75} = 0.6667$$

Thus, the unconditional variance of the process is approximately **0.6667**.

The correct answer is: **0.6667**.

Question 3 (1 point)
An AR(1) process is Y(t) = 2 + 6Y(t-1) + e(t). What is the unconditional mean of the process?

Question 3 options:

A finite unconditional mean does not exist and process is not stationary

6

4

0

For an AR(1) process to have a finite unconditional mean, it needs to be stationary. A necessary condition for stationarity is that the absolute value of the coefficient of $Y(t-1)$ (denoted as ϕ) must be less than 1.

The given AR(1) process is:

$$Y(t) = 2 + 6Y(t-1) + e(t)$$

Here, $\phi = 6$, which is greater than 1. When ϕ is greater than 1, the process is non-stationary, and the unconditional mean does not exist.

Therefore, the correct answer is:

A finite unconditional mean does not exist and the process is not stationary.

Question 4 (1 point)
What is the unconditional mean of the following MA(2) process: Y(t) = 5 + 0.1e(t-1) + 0.2e(t-2) + e(t)

Question 4 options:

5

25

0.1

50

For a Moving Average (MA) process, the unconditional mean is simply the constant term in the process equation. This is because the error terms ($e(t)$) have a mean of zero, and the coefficients of the error terms do not affect the mean.

The given MA(2) process is:

$$Y(t) = 5 + 0.1e(t-1) + 0.2e(t-2) + e(t)$$

Here, the constant term is 5. The error terms ($e(t)$, $e(t-1)$, and $e(t-2)$) have a mean of 0, so they do not contribute to the unconditional mean.

Therefore, the unconditional mean of the process is **5**.

The correct answer is: **5**.

Question 5 (1 point)
Which of the following is true of AR(p) and MA(q) processes?

Question 5 options:

Both AR(p) and MA(q) processes are always covariance stationary

Both AR(p) and MA(q) processes are never covariance stationary

AR(p) processes may or may not be covariance stationary but MA(q) is always covariance stationary

Both AR(p) and MA(q) processes may or may not be covariance stationary depending on coefficients, p and q.

The correct answer to this question depends on the nature of AR(p) and MA(q) processes:

- **AR(p) (AutoRegressive) processes**: Whether an AR(p) process is covariance stationary depends on the values of the coefficients. The process is covariance stationary if the roots of the characteristic equation lie outside the unit circle (i.e., the absolute value of the autoregressive coefficients must be less than 1 for stationarity). So, AR(p) processes **may or may not** be covariance stationary depending on the coefficients.
- **MA(q) (Moving Average) processes**: MA(q) processes are always covariance stationary because they are linear combinations of error terms, which are assumed to have constant variance and mean. No specific condition is required for covariance stationarity in MA(q) processes.

Therefore, the correct answer is:

AR(p) processes may or may not be covariance stationary but MA(q) is always covariance stationary.

Question 6 (1 point)
What is a characteristic of ACF for an AR(1)?

Question 6 options:

Exponentially but smoothly decaying in absolute value as lag k increases

Exponentially but smoothly increasing in absolute value as lag k increases

Slowly increasing with lag k up to a certain point, then sharply dropping

Slowly decreasing with lag k up to a certain point, then sharply dropping

For an **AR(1)** process, the autocorrelation function (ACF) exhibits a characteristic pattern:

- The ACF **decays exponentially** as the lag k increases. The rate of decay depends on the value of the autoregressive coefficient (ϕ) of the AR(1) process. As the lag increases, the autocorrelations get smaller and eventually approach zero smoothly.

Thus, the ACF for an AR(1) process **decays exponentially but smoothly in absolute value as lag k increases**.

The correct answer is:

Exponentially but smoothly decaying in absolute value as lag k increases.

Question 7 (1 point)
What is a characteristic of the PACF for AR(3)?

Question 7 options:

Slowly decreasing in absolute value up to k = 36, then sharply falls towards 0 in absolute value

None of above

Absolute value takes smooth exponential decay towards 0 as lag k increases

Slowly decreasing in absolute value up to k = 3, then sharply falls towards 0 in absolute value

For an **AR(3)** process, the Partial Autocorrelation Function (PACF) has a distinctive characteristic:

- The PACF for an AR(p) process will show significant partial autocorrelations for the first p lags (in this case, $p = 3$). After that, the PACF will **drop off sharply** to zero for higher lags. This is because the AR process is fully captured by the first p lags, and further lags do not add new information.

Therefore, the PACF for an AR(3) process will show significant values for the first 3 lags and then sharply drop toward zero for lags greater than 3.

The correct answer is:

Slowly decreasing in absolute value up to $k = 3$, then sharply falls towards 0 in absolute value.

Question 8 (1 point)
What is an ARMA(1,2) model?

Question 8 options:

Product of AR and MA models of arbitrary order to be determined by automated modelling

Simple sum of AR(2) and MA(1) models

Simple sum of AR(1) and MA(2) models

Model made up of machine learning algorithm with AR and MA models as inputs

An **ARMA(1,2)** model is a combination of an autoregressive (AR) model of order 1 and a moving average (MA) model of order 2. The general form of an ARMA(1,2) model is:

$$Y(t) = \phi_1 Y(t-1) + e(t) + \theta_1 e(t-1) + \theta_2 e(t-2)$$

Where:

- ϕ_1 is the coefficient of the AR(1) part,
- θ_1 and θ_2 are the coefficients of the MA(2) part,
- $e(t)$ represents the error terms.

Thus, the ARMA(1,2) model is the **sum of an AR(1) model and an MA(2) model**.

The correct answer is:

Simple sum of AR(1) and MA(2) models.

Question 9 (1 point)
What is the use of ACF and PACF for a time series?

Question 9 options:

We can always use ACF to exactly determine the type of model and PACF to determine the order for both AR and MA

We can always use PACF to exactly determine the type of model and ACF to determine the order for both AR and MA

We can always use both in conjunction to exactly determine the type of model and order exactly

To narrow down possibilities and form educated guesses as to the type and order of time series model (AR/MA/ARMA/ARIMA), and in some simple cases, allow us to directly chose the model type and order

The **Autocorrelation Function (ACF)** and **Partial Autocorrelation Function (PACF)** are valuable tools in identifying and selecting appropriate time series models, but they are not always definitive in determining the exact model type and order.

- **ACF** helps in identifying whether a Moving Average (MA) component is present and estimating the order of the MA model.
- **PACF** helps in identifying whether an Autoregressive (AR) component is present and estimating the order of the AR model.

However, these functions are typically used to **narrow down possibilities** and guide model selection. In some simple cases, they can directly suggest the model type and order (such as when the ACF or PACF cuts off after a few lags), but more complex models may require further diagnostic checks or automated model selection methods.

The correct answer is:

To narrow down possibilities and form educated guesses as to the type and order of time series model (AR/MA/ARMA/ARIMA), and in some simple cases, allow us to directly choose the model type and order.

Question 10 (1 point)
What is the ARIMA model?

Question 10 options:

None of above

Recursively estimate AR or MA models on a raw variable, taking the result of each step as the input to the next

Take one or more differences on variable to make it stationary, then estimate an ARMA model on (possibly differenced) variable

Estimate an ARMA model on raw variable

The **ARIMA** model stands for **AutoRegressive Integrated Moving Average** and is used for modeling time series data that may not be stationary. It involves the following steps:

1. **Differencing the series** one or more times to make it stationary (this is the "Integrated" part, represented by I).
2. **Estimating an ARMA model** (AutoRegressive Moving Average) on the differenced series.

In other words, ARIMA is used when the time series is non-stationary, and it involves differencing the data to remove trends or other non-stationary components before applying ARMA modeling.

Therefore, the correct answer is:

Take one or more differences on a variable to make it stationary, then estimate an ARMA model on (possibly differenced) variable.

Question 11 (1 point)
Residuals in a time series estimation (AR, MA, ARMA, ARIMA) should have what properties?

Question 11 options:

Zero autocorrelation

Constant variance

Mean = 0

All of the above

In time series modeling (whether AR, MA, ARMA, or ARIMA), the **residuals** (the difference between the observed values and the model's predicted values) should exhibit the following properties for the model to be considered well-fitted:

1. **Zero autocorrelation**: The residuals should not show any autocorrelation, meaning there should be no predictable patterns left in the residuals. This indicates that the model has captured the dependencies in the data.

2. **Constant variance** (homoscedasticity): The residuals should have a constant variance, meaning that their spread should not change over time.
3. **Mean = 0**: The residuals should have a mean of zero, implying that the model is unbiased and is neither consistently overestimating nor underestimating.

Therefore, the correct answer is:

All of the above.

Question 12 (1 point)
A very extremely slowly decaying ACF without any seasonal patterns is a clear and obvious indication of which of the following?

Question 12 options:

Underlying variable is close to stationary

Seasonality in the data

Long term directional trend in the data; data is not stationary and likely requires one or more differences

Underlying variable is white noise

A **very slowly decaying ACF** without any seasonal patterns typically indicates that the time series has a **long-term directional trend** and is **non-stationary**. In such cases, the data likely requires **differencing** to remove the trend and make the series stationary. This is a classic sign of non-stationarity because, in stationary processes, the ACF usually decays quickly.

The correct answer is:

Long term directional trend in the data; data is not stationary and likely requires one or more differences.

Question 13 (1 point)
An MA(1) process is Y(t) = 10 + 0.5e(t-1) + e(t). What is the autocorrelation between Y(t) and Y(t-10)?

Question 13 options:

0

0.5

None of above

0.6

In a **Moving Average process of order 1** (MA(1)), the autocorrelation function (ACF) has a specific behavior:

- For an MA(1) process, autocorrelations are only present at lag 1.
- For lags greater than 1, the autocorrelations drop to zero.

The given MA(1) process is:

$$Y(t) = 10 + 0.5e(t-1) + e(t)$$

Since we are asked for the autocorrelation between $Y(t)$ and $Y(t-10)$, which is a lag of 10, the autocorrelation will be **0** because the autocorrelation function for an MA(1) process only extends to lag 1.

Therefore, the correct answer is: **0**.

Question 14 (1 point)
An AR(1) process is Y(t) = 1 + 0.1Y(t-1) + e(t). What is the

autocorrelation between Y(t) and Y(t-5)?

Question 14 options:

0.1

0.1 to the power of 5

1

(1-0.1) to the power of 5

For an **AR(1) process**, the autocorrelation between $Y(t)$ and $Y(t-k)$ (lag k) is given by the formula:

$$\rho(k) = \phi^k$$

Where:

- ϕ is the coefficient of $Y(t-1)$,
- k is the lag (in this case, $k = 5$).

The given AR(1) process is:

$$Y(t) = 1 + 0.1Y(t-1) + e(t)$$

Here, $\phi = 0.1$, so the autocorrelation between $Y(t)$ and $Y(t-5)$ is:

$$\rho(5) = (0.1)^5$$

Therefore, the correct answer is: **0.1 to the power of 5**.